CS543 Web Services — Assignment 2

Multi-Agent AI Workspace — Services, Contracts & Schema

Capabilities

• User Authentication & Access
• Workspace Management
• Task Management
• AI Task Orchestration
• Specialized Agent Execution
• Shared Agent Context
• Review & Validation
• Artifact & Execution Management
• History & Monitoring
• Final Result Delivery

Service ownership

Service	Owner	Owned data
Identity & Access	Harendra Godara	users, auth_sessions, roles/access
Workspace & Task	Harendra Godara	workspaces, members, tasks, dependencies, workflow_runs
AI Agent	Mohit Tailor	agents, agent_runs, messages, agent_context
Execution & Artifact	Anurag Sharma	artifacts, reviews, execution_logs, test_runs
Frontend/UI	Harshita Rathore	presentation only

Central operation — runTask

Inputs: authenticated user/workspace context, task description/reference and execution options. Outputs: workflow run status, progress, final reviewed result and artifact/log references.

Errors: 401, 403, 404, 409, 422, 429 and 503 as appropriate. Hidden implementation details include SQL, database tables, internal IDs, agent-selection algorithm, private prompts/context, credentials, retry mechanics and transaction boundaries.

Contract examples

Operation	Input	Output	Errors
loginUser	email,password	token,user	400,401
createTask	workspace,title,description	task	401,403,404,422
runTask	task,options	workflow run	401,403,404,409,422,429,503
executeTask	task,context	agent output	404,409,429,503
saveOutput	run,output	artifact ref	400,503

Five Service Properties

Property	Result	Reason
Reachable	Pass	API endpoint/contract exists.
Self-contained	Pass	Own persistent data.
Contract	Pass	Input/output/errors published.
Independent	Mostly pass	Can be deployed/configured separately.
Loosely coupled	Pass	API calls, no shared databases.

Schema boundary: each persistent table has one owner; cross-service identifiers are API references and cross-service database foreign keys are avoided.